

# Demonstrating PopCore: Personal Fabrication of 3D Foamcore Models for Professional High-Quality Applications in Design and Architecture

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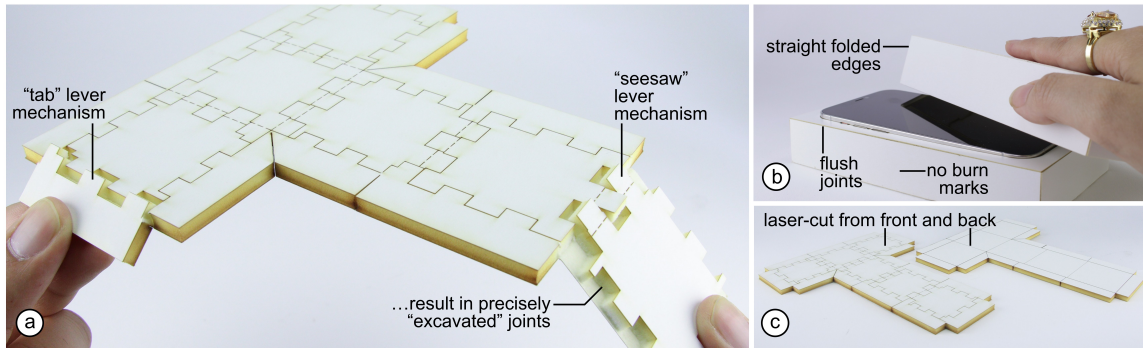
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**Figure 1:** We demonstrate *PopCore*, a fabrication technique that laser-cuts *high-quality* 3D models from paper-foam-paper sandwich materials. (a) Its key elements are two laser-cut lever mechanisms that allow users to break off surrounding residue material, thereby “excavating” joints with *very high* precision, giving models (b) a look that is cleaner and more homogeneous than any prior fabrication technique. (c) *PopCore* achieves this by laser cutting from both the front and the back. *PopCore*’s clean appearance allows personal fabrication to tackle fields that require a *professional* look, in particular industrial design, architecture, and high-end packaging design.

## ABSTRACT

*PopCore* is a fabrication technique that laser-cuts 3D models from paper-foam-paper sandwich materials. Its key elements are two laser-cut lever mechanisms that allow users to break off surrounding residue material, thereby “excavating” joints efficiently and with very high precision, which *PopCore* produces by laser cutting from the top and bottom. This produces flush joints, folded edges that are perfectly straight, and no burn marks—giving models a homogeneous, clean look. This allows applying personal fabrication

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to new fields, including industrial design, architecture, and packaging design, that require a visual finish beyond what traditional personal fabrication delivers. We demonstrate the algorithms and a software tool that generates PopCore automatically. Our user study participants rated PopCore models significantly more visually appealing (7.9/9) than models created using techniques from the related work (4.7/9 and 2.3/9) and suitable for presentation models (11/12 participants), products (10/12 participants) and high-end packaging (10/12 participants).

## CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

## KEYWORDS

Personal fabrication, Laser cutting, Rapid prototyping, Manual assembly

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## 1 INTRODUCTION

While tools automating the personal fabrication of 3D models have been very successful with makers (*CoilCAM* [7], *Makers' marks* [20], *FusePrint* [22]), these tools have not yet found their way into professional design disciplines, such as industrial design, architecture, or packaging design. Thus, even though the use of fabrication machinery in these fields goes back much further than the maker movement, designers in these fields continue to design and engineer their models using traditional fabrication techniques and software tools that provide full control (e.g., AutoCAD [4]) while forgoing the speed and convenience that could be achieved with the recent maker tools.

One of the key reasons for this is that the professional design disciplines demand high-quality finish ([9], [6], [19], [16]), while it is only *desirable* for makers. Thus, while laser cutting software addressing makers tends to create cross joints [14] and box joints [5], [3]—these, however, produce visual artifacts so prominent that they distract from the actual design intent [15]. Professional designers and architects instead tend to focus on stacking plates [11], (manually cut) miter joints [11], and in some cases butt joints [13]. These in turn require substantial manual skill, resulting in untrained users - such as makers - producing results that are much less sturdy—thus affording mostly only decorative objects. As a result, we see two separate fields—makers on one side and professional designers on the other side—each using their own techniques and technologies.

Recent work has made a promising step towards bringing the two together: “HingeCore” [2] produces box joint constructions, which are reasonably sturdy and afford fast assembly, yet offer a

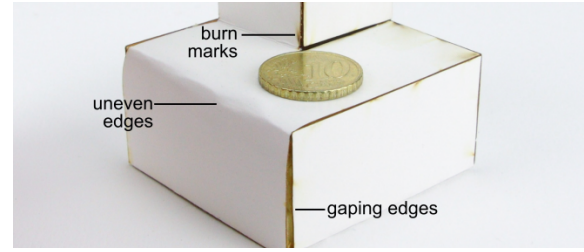


Figure 2: Artifacts introduced by HingeCore.

*somewhat* clean look: HingeCore folds paper-foam-paper “foam-core” sandwich material ([12], [17]) into 3D structures, but hides the box joints behind one of the paper layers [2].

Unfortunately, HingeCore introduces its own set of artifacts in the form of gaping and uneven edges (Figure 2). In our user study, participants found the resulting quality suitable for *internal* prototyping, but not for creating customer-facing use cases, such as presentation models and products for industrial design, architecture, and packaging.

In this paper, we address these issues with a new take on laser-cut foamcore. We introduce *PopCore*. As illustrated by Figure 1, PopCore is a fabrication technique that produces 3D models with an artifact-free look. (a) PopCore’s key design elements are two laser-cut lever mechanisms that allow users to break off surrounding residue material, thereby “excavating” joints efficiently and with very high precision. (b) This results in flush joints, folded edges that are perfectly straight, while avoiding burn marks, resulting in 3D models with a homogeneous, clean look. The resulting high-quality look allows PopCore to address fields that require high-end visual finish, including industrial design, architecture, and, as illustrated here, packaging design. (c) We demonstrate an algorithm and a software tool that generate PopCore automatically by laser cutting not only from the top, but also from the bottom.

We refer to our full paper [1] for implementation details, alternative materials, applications, and user study.

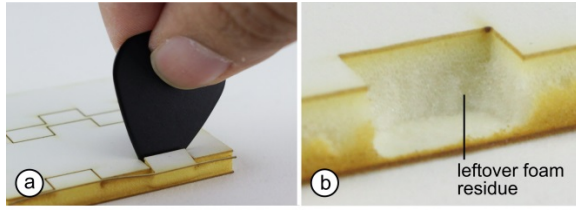
## 2 LASER-CUT SANDWICH MATERIALS

Foldem [18] introduces laser cutting sandwich material halfway, a key technique that allows creating artifacts with both rigid and flexible properties or embed electrical traces into geometry (LaCir [8]) in one go. HingeCore builds on this by laser cutting foamcore halfway thereby producing “finger hinges” [2]. Finger hinges speed up assembly by allowing users to fold models, while simultaneously allowing for sturdy construction based on box joints.

In the context of this paper, the key benefit of HingeCore is that it introduces hidden box joints, thereby introducing the *vision* of sturdy 3D models with an artifact-free finish.

Unfortunately, as illustrated by Figure 2, HingeCore introduces its own set of artifacts: visible edges tend to get burnt, edges appear uneven, and joints tend to gape.

Figure 3 illustrates why joints tend to gape: (a) picking out the material blocks between fingers (aka “residue”), (b) commonly leaves bits of foam behind. When assembling the model, the leftover foam prevents fingers from entering all the way, resulting in the

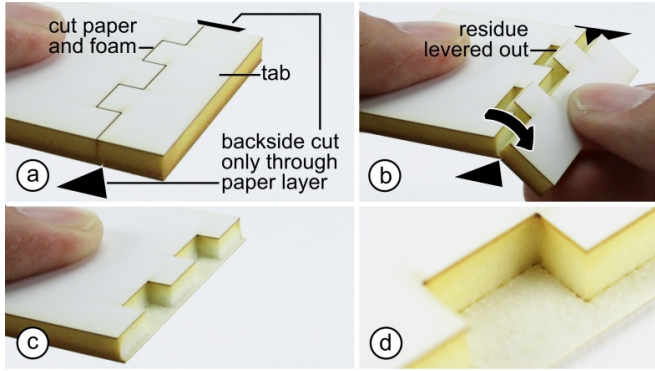


**Figure 3: Why HingeCore tends to gape: (a) Pulling residue out, (b) frequently leaves bits of foam behind.**

aforementioned gapping. The heat-shrinking polystyrene mentioned in [2] faces the same issue—here it is the shrunk material left behind in the cavities.

### 3 POPCORE

The key elements of PopCore are two lever mechanisms. Their main purpose is to pull out residue easily and cleanly.

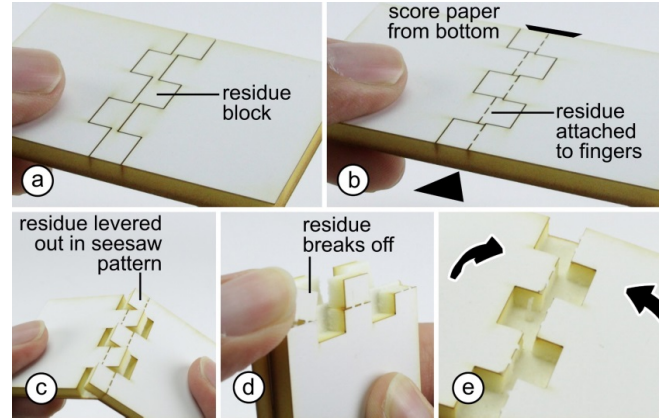


**Figure 4: PopCore enables complete removal of residue by creating (a) tabs along the edge that connect to the residue, (b) lever it out along a shallow cut at the bottom and (c) break off.**

Figure 4a illustrates the tab lever mechanism. Rather than producing isolated blocks of residue (Figure 3), tab mechanisms connect the material to be removed to material located outside the part. Generally, all fingers along an edge form a single such tab. (b) During assembly, these tabs act as levers. Jerking the tab downwards causes (c) the residue to be pulled upwards, thus ripping it out of its cavity—until the tab finally breaks off. (d) Since the cohesion of the foam is higher than the adhesion to the paper layer, the foam stays in one piece, leaving a cavity that is perfectly clear ready to be joined with its counterpart.

The key element that creates the lever is a shallow straight cut across the bottom paper layer, hinted at in Figure 4a by a pair of black triangles and revealed in (c). This cut forms the fulcrum of the lever. The bottom paper layer is inaccessible from the top, as it is partially hidden behind the fingers, which is why PopCore creates them by cutting from below.

Figure 5 illustrates the seesaw lever mechanism. It allows creating joints located along a folded edge.



**Figure 5: (a) In order to remove these blocks of residue, (b) Seesaw levers attach blocks of residue to their respective opposing finger using a perforated edge. (c) Users lever out the residue in a “seesaw” pattern by folding down, and (d) breaks it off from the fingers. (e) Resulting in a residue free joint ready to be folded.**

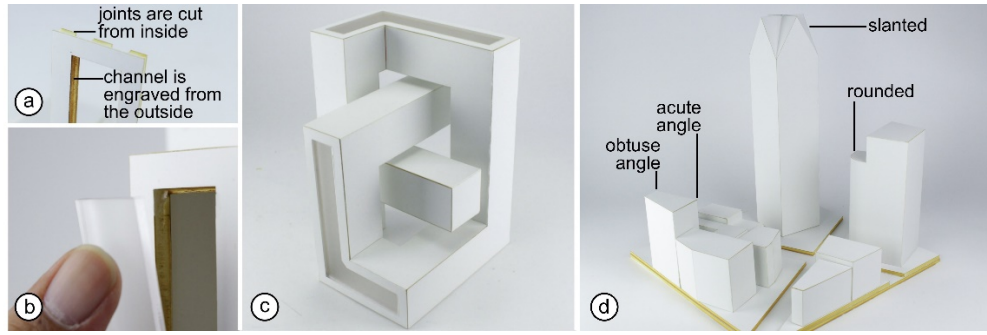
As a side effect of cutting from both sides (Figure 6), PopCore naturally allows placing arbitrary features on either side. (a) This allows engraving a recessed press-fit channel. (b) The engraved channel allows press-fitting the frosted acrylic panels that act as diffusers for (c) a lamp which we replicate from [2]. (d) PopCore not only supports rectilinear geometry, but also slanted and rounded geometry. Rounded edges are created using the same techniques as HingeCore.

The concept of fabricating presentation models and products for sale is a powerful one. In Figure 7, we think this idea through. PopCore allows the shown model to ship in a flat-pack envelope. Upon arrival, PopCore’s lever mechanisms allow users to “excavate” parts quickly and cleanly. Parts then assemble efficiently. The assembled 3D models, finally, are free of gapping and burn marks as one would expect from a product.

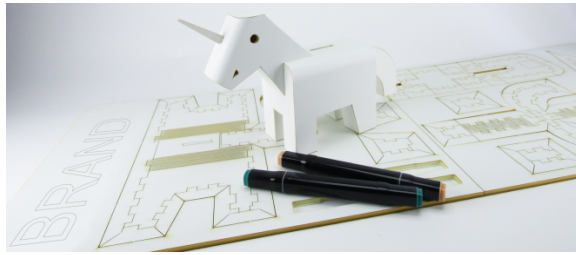
Figure 8 illustrates the cutting plan created by PopCore, at the example of a 5x5x5 cm cube. The laser cutting process takes place in two passes. First, the laser cutter executes the (a) red (cut through), (b) green (cut top paper layer), and (c) yellow (crease top paper layer) lines. To simplify laser cutting from both sides PopCore also creates (d) a rectangular corner as part of the layout.

Users now flip the layout and (f) align the rectangular corner against the alignment aids commonly found in laser cutters along the top and left edges (e.g. seen in Trotec [21], and Glowforge [10]). This process produces precision substantially beyond what PopCore requires. (g) PopCore lays out the second side under this assumption, so cutting can start right away without further calibration activities. This allows aligning the mirrored cut-ting plan with its physical counterpart. The laser cutter now creates the finger hinges shown in magenta by cut-ting “half-way”, i.e., through the top paper layer and the foam layer.

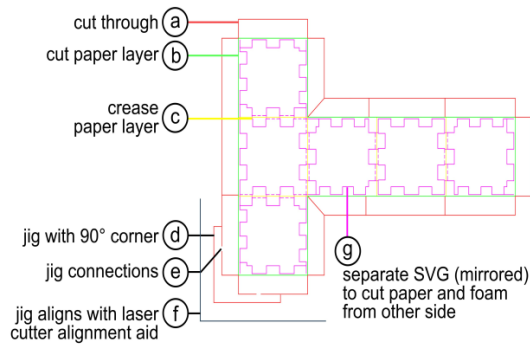
(e) Small breaks in the red outline keep the layout connected to the jig during flipping but allow for easy removal later. Settings to



**Figure 6: PopCore supports engraving and cutting from either side and also supports rectilinear, rounded, and slanted geometry.**



**Figure 7: Envisionment of a product fabricated using PopCore**



**Figure 8: The laser cutter fabricates PopCore by executing the first half of the cutting plan (a-c). Users then flip the workpiece in the laser cutter and align it by pushing (d) the 90° jig into (f) the alignment aid of the laser cutter. (g) Magenta lines are now cut.**

laser-cut at different layer depths are machine specific and based on [2].

## 4 CONCLUSION

In this paper, we demonstrated PopCore, a set of novel fabrication techniques and geometric structures that increase the visual quality of laser-cut foamcore models to the level of professional design disciplines, such as industrial design, architecture, and packaging design. The key is the underlying concept of an embedded, laser-cut lever mechanism, which produces very high quality, while being fast and enjoyable to use.

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